

UNIT 2: MOVEMENT OF MOLECULES

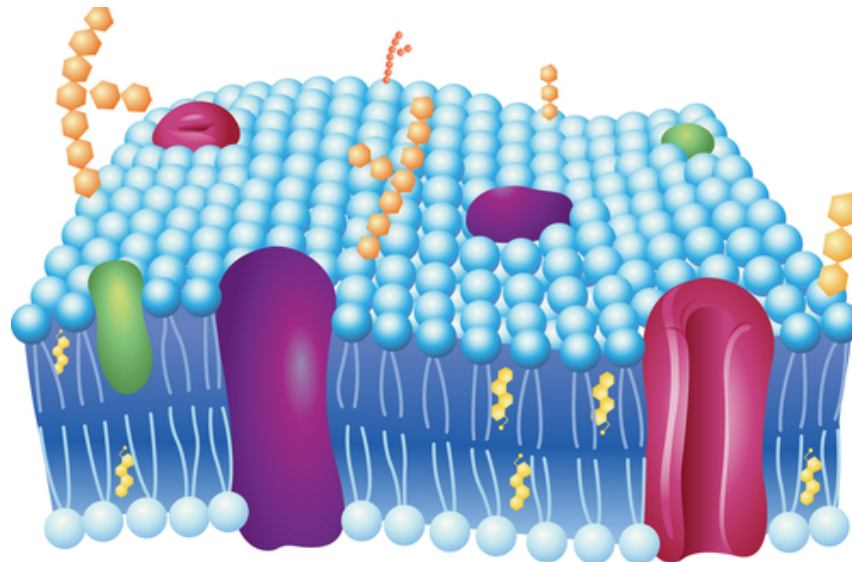
Biology 9

Mr. Queenan

Text - Ch 8.3 (p 260-273)

Membrane Structure & Function

- Phospholipids, the key ingredient of biological membranes, spontaneously self-assemble into simple membranes
- ▣ The formation of membrane-enclosed collections of molecules was a critical step in the evolution of the first cells.



Membrane Structure & Function

- There are other molecules that are embedded in the phospholipid bilayer, and are relatively uniform across most animal cell membranes
 - ▣ Cholesterol and other lipid molecules
 - ▣ Glycoproteins and glycolipids
 - ▣ Proteins
- These play a part in the fluidity of the membrane, and both the passive and active permeability of the membrane.
- Fluid Mozaic Model

Membrane Bound Lipids

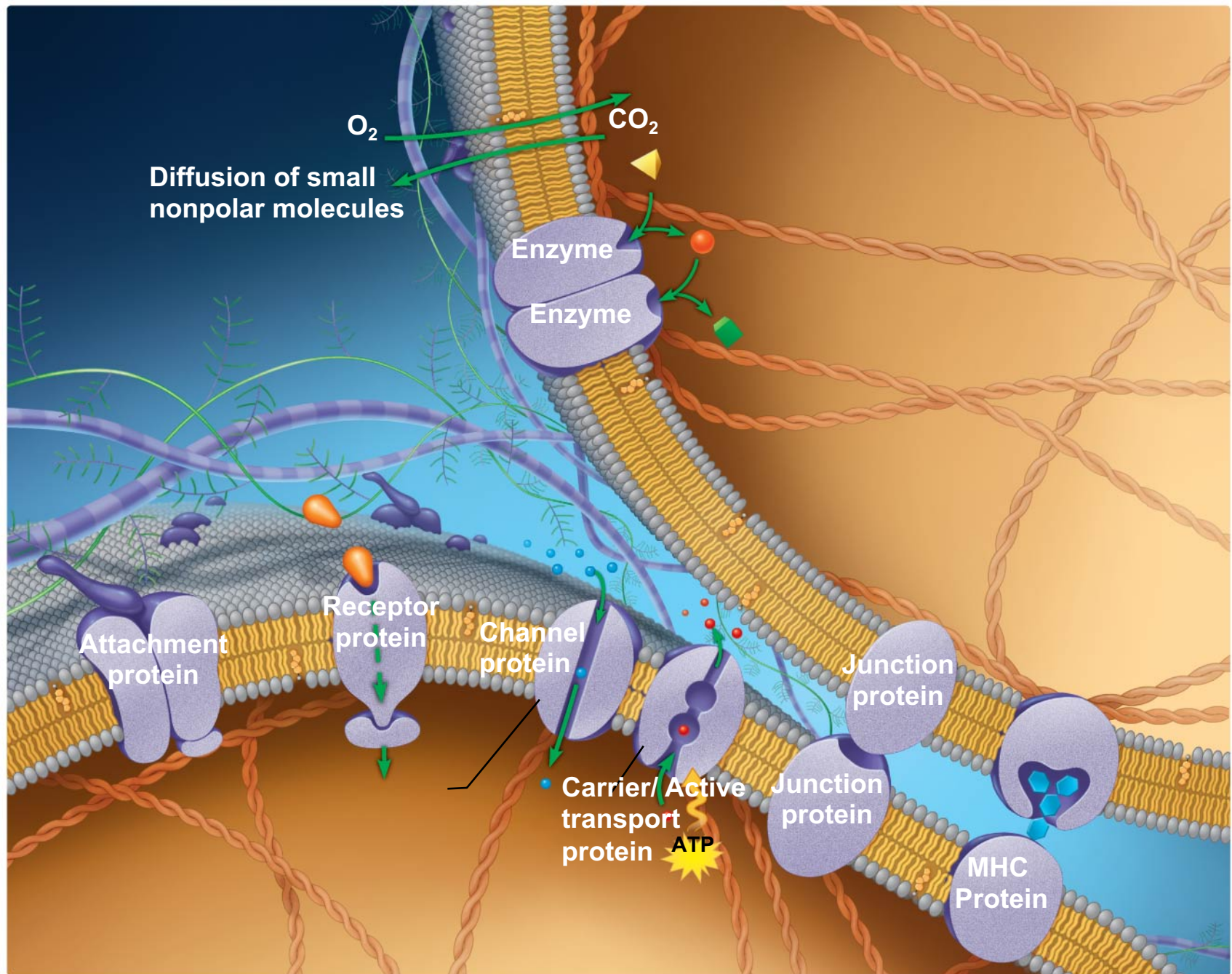
- Cholesterol and phospholipids
 - ▣ Help create membrane fluidity
 - ▣ Stop membrane from becoming too fluid at high temps; too rigid at cold temps
- Individual molecules can flip from extracellular monolayer to intracellular monolayer.

Membrane Bound Carbohydrates

- Glycoproteins and glycolipids
 - Phospholipids or proteins that have a sugar chain attached
 - Carbohydrate chains only occur on the outside surface of cell, making cell membrane “asymmetrical”
 - Main purpose of these molecules:
 - Cell-to-cell adhesion
 - Reception of signaling molecules
 - Cell-to-cell signaling
 - Diversity of carbohydrate chains leads to cellular “fingerprints”

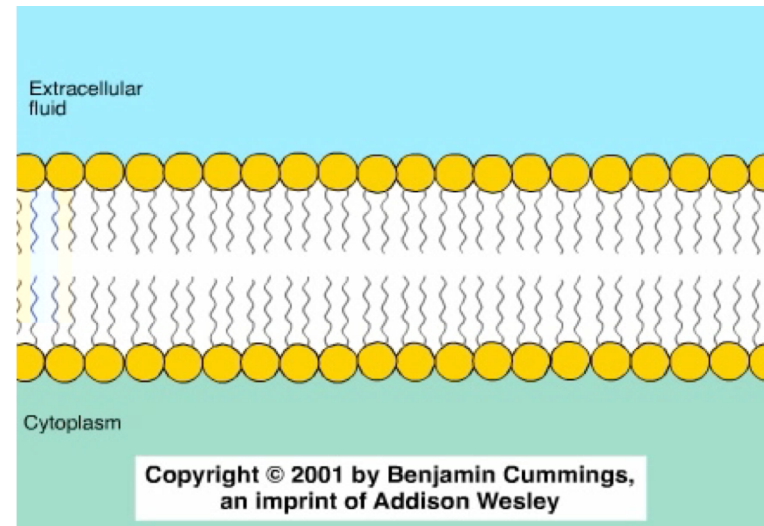
Membrane Bound Proteins

- The proteins found in a membrane can differ depending on type of cell, or tissue system.
- 6 main types of proteins, each with its own function
 - ▣ Channel Protein – allows passage of molecules or ions
 - ▣ Carrier Protein – selectively interacts w/ specific molecule
 - ▣ Cell Recognition Protein (MHC) – immunity/organ transplant
 - ▣ Receptor Protein – shaped so a molecule can bind to it
 - ▣ Enzymatic Protein – catalyzes a specific reaction
 - ▣ Junction Protein – joins cells to allow a tissue to function



Membrane Permeability

- Plasma membrane of cells is selectively permeable
 - Allows in some substances while keeping other out.
 - Some molecules can freely move across the membrane without needing to expend any energy
 - Hydrophobic molecules are similar to inside of the bilayer, and can move across at no energy cost
 - Polar molecules require energy input to move across membrane



Concentration Gradient

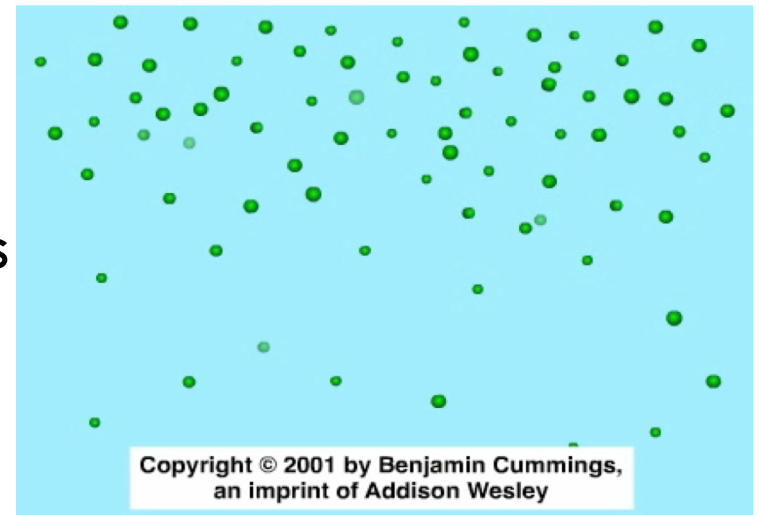
- Most molecules follow a concentration gradient
 - Move from areas of higher concentration to lower until equilibrium is reached
 - Ex. Oxygen and Carbon Dioxide in the cell.
 - Oxygen is always flowing:
 - Into the cell because oxygen is being used
 - Carbon dioxide is always flowing:
 - Out of the cell because it is building up as a waste product of reactions

Passive Transport - Diffusion

- **Diffusion** – Movement of molecules from a higher to lower concentration until equilibrium is achieved and molecules are distributed equally.

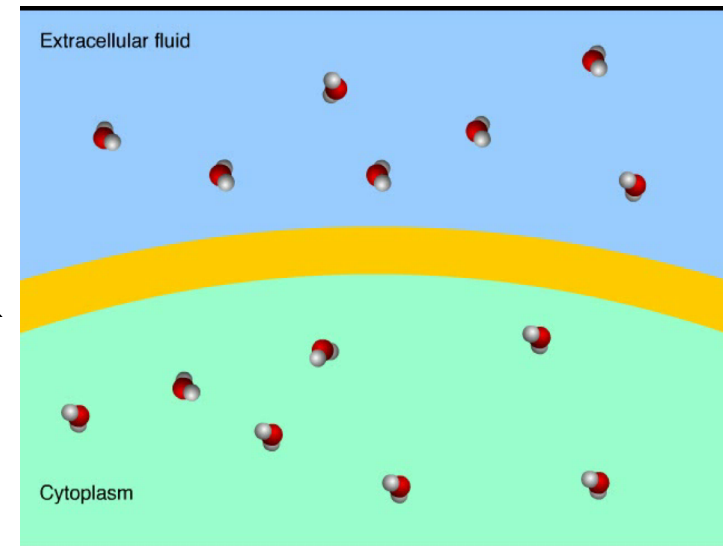


- Random process
- In cells, only certain molecules can enter and exit cells by diffusion
 - Gases – small and nonpolar



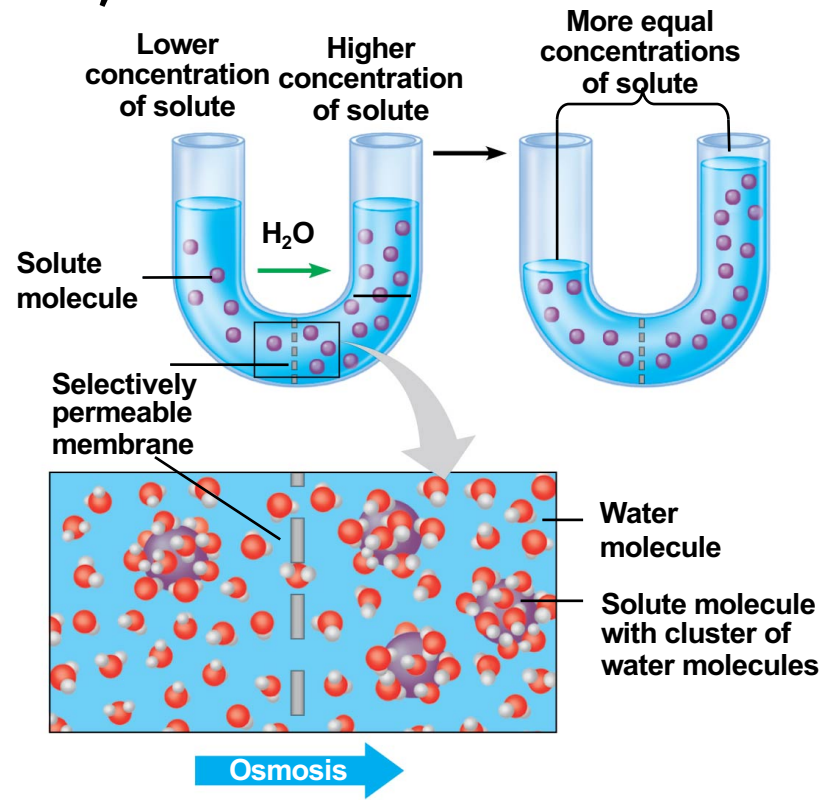
Passive Transport - Osmosis

- **Osmosis** – The diffusion of water across a semipermeable membrane from high to low concentration
- Water flows from areas of high osmotic pressure to low osmotic pressure
 - This is the process by which water is absorbed by the kidneys in the body &
 - Taken up by capillaries in tissues &
 - How water crosses membranes



Osmosis

- If a membrane, permeable to water but not to a solute, separates two solutions with different concentrations of solute, water will cross the membrane, moving down its own concentration gradient, until the solute concentration on both sides is equal.



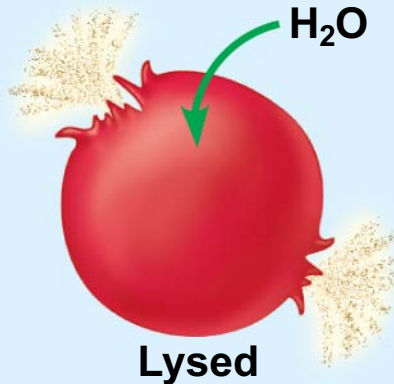
Tonicity

- **Tonicity** is a term that describes the ability of a surrounding solution to cause a cell to gain or lose water.
- The tonicity of a solution mainly depends on its concentration of solutes relative to the concentration of solutes inside the cell.
 - ▣ In an **isotonic** solution, the concentration of solute is the same on both sides of a membrane, and the cell volume will not change.
 - ▣ In a **hypotonic** solution, the solute concentration is lower outside the cell, water molecules move into the cell, and the cell will expand and may burst.
 - ▣ In a **hypertonic** solution, the solute concentration is higher outside the cell, water molecules move out of the cell, and the cell will shrink.

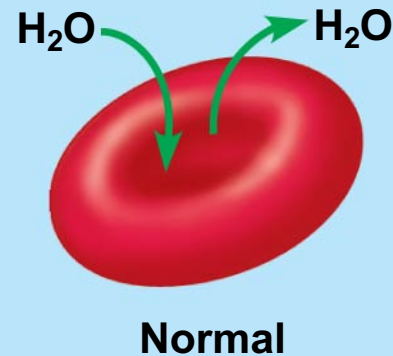
Tonicity

Animal cell

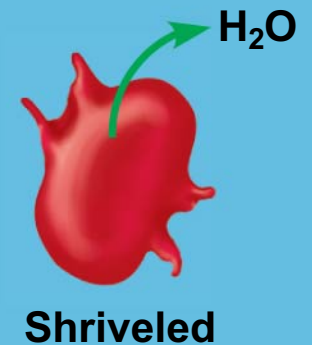
**Hypotonic solution
(lower solute levels)**



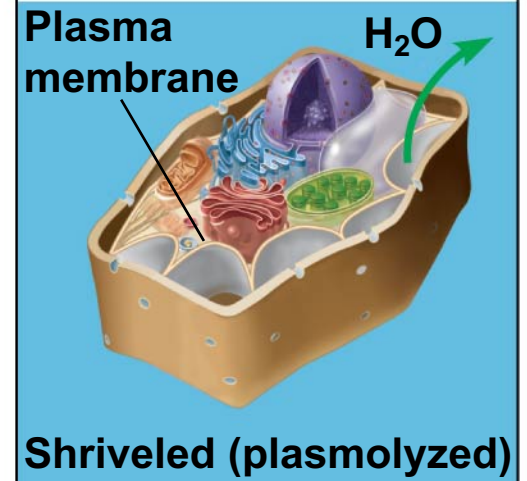
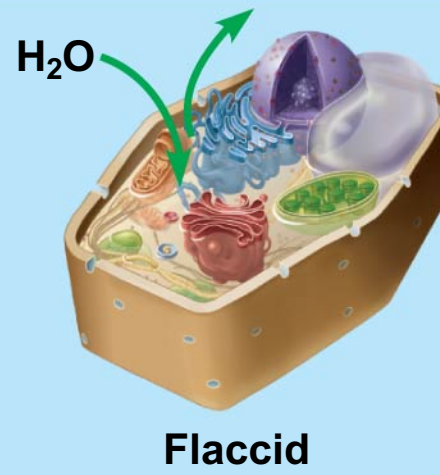
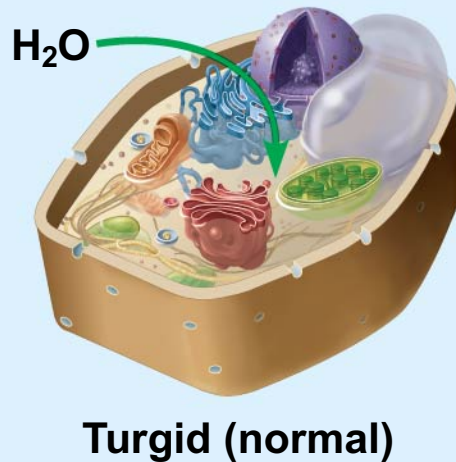
**Isotonic solution
(equal solute levels)**



**Hypertonic solution
(higher solute levels)**



Plant cell

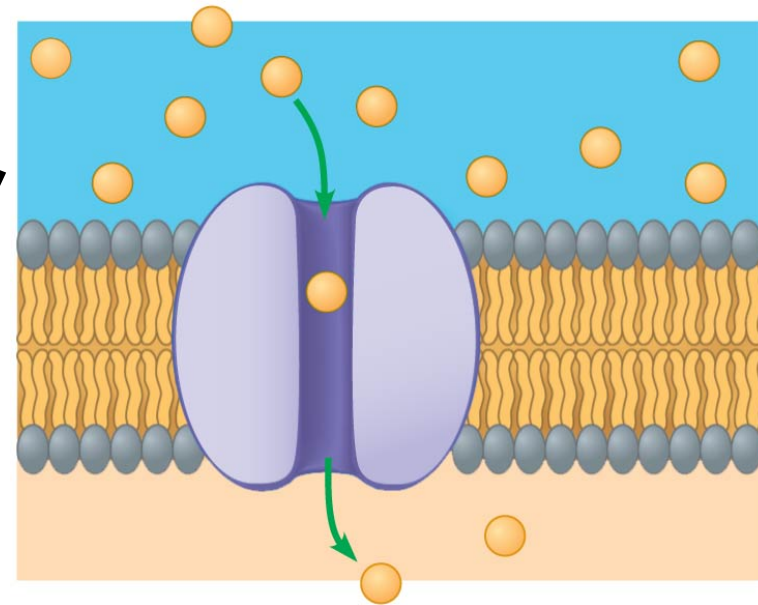


Facilitated Diffusion

- Hydrophobic substances easily diffuse across a cell membrane.
- However, polar or charged substances do not easily cross cell membranes and, instead, move across membranes with the help of specific transport proteins in a process called **facilitated diffusion**, which
 - ▣ does not require energy and
 - ▣ relies on the concentration gradient.

Facilitated Diffusion

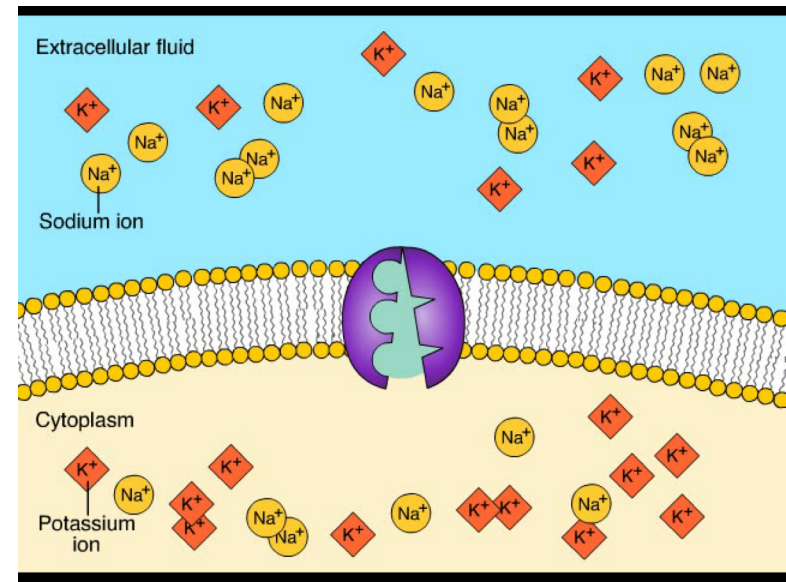
- Some proteins function by becoming a hydrophilic tunnel for passage of ions or other molecules.
- Other proteins bind their passenger, change shape, and release their passenger on the other side.
- In both cases, the transport protein helps a specific substance diffuse across the membrane down its concentration gradient and thus requires no input of energy.



Aquaporin is specific type of protein identified in the rapid diffusion of water into or out of a cell

Active Transport

- In active transport, a cell must expend energy to move a solute *against* its concentration gradient.
- The energy molecule ATP supplies the energy for most active transport.
 - ▣ Proteins involved in active transport often called pumps
 - ▣ Sodium-Potassium Pump most famous in a cell



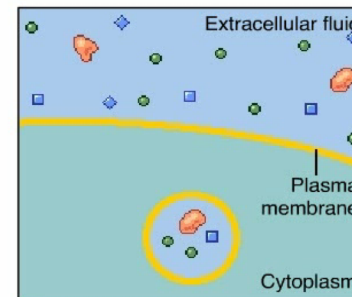
Bulk Transport of Molecules

- A cell uses two mechanisms to move large molecules across membranes.
 1. **Exocytosis** is used to export bulky molecules, such as proteins or polysaccharides.
 2. **Endocytosis** is used to take in large molecules.
- In both cases, material to be transported is packaged within a vesicle that fuses with the membrane.

Exocytosis

- Intracellular vesicle fuses with plasma membrane and secretion of molecule occurs
- Used for export of items such as:

- Hormones
- Neurotransmitters
- Digestive enzymes



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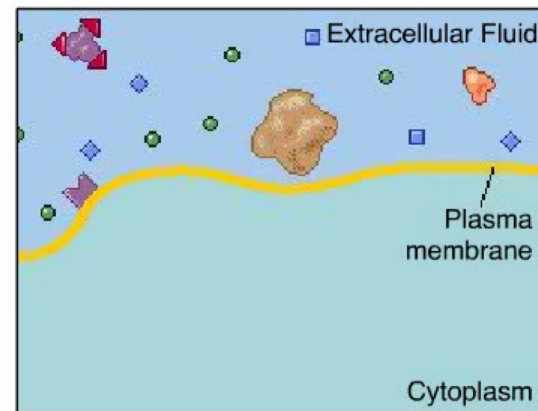
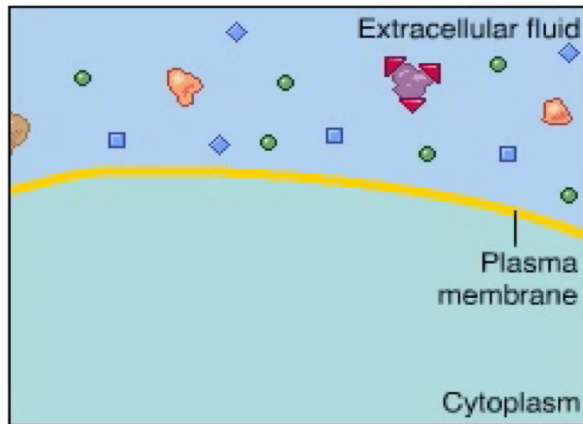
- Often produced by Golgi body in the cell
- Vesicle becomes part of plasma membrane upon fusing

Endocytosis

- There are two kinds of endocytosis.
 1. **Phagocytosis** is the engulfment of a particle by the cell wrapping cell membrane around it, forming a vacuole.
 1. **Pinocytosis** when engulfment of a liquid
 2. **Receptor-mediated endocytosis** uses membrane receptors for specific solutes. The region of the membrane with receptors pinches inward to form a vesicle.
- Receptor-mediated endocytosis is used to take in cholesterol from the blood.

Endocytosis

Pinocytosis



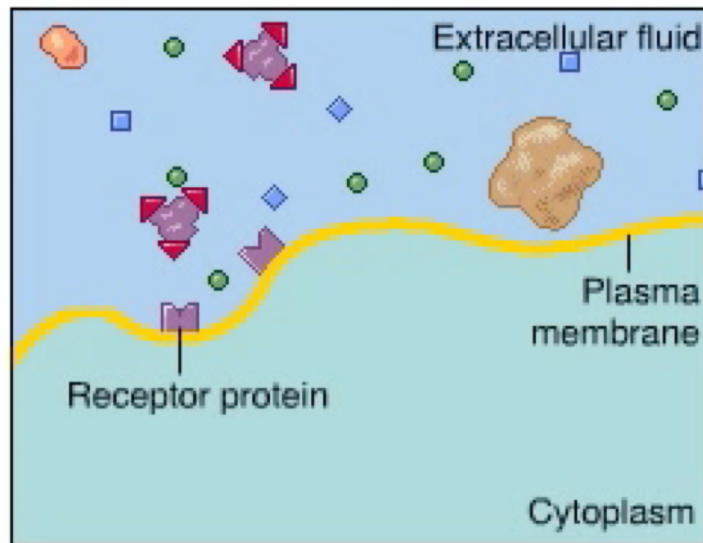
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Phagocytosis

Endocytosis

Receptor-mediated Endocytosis



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To Review

